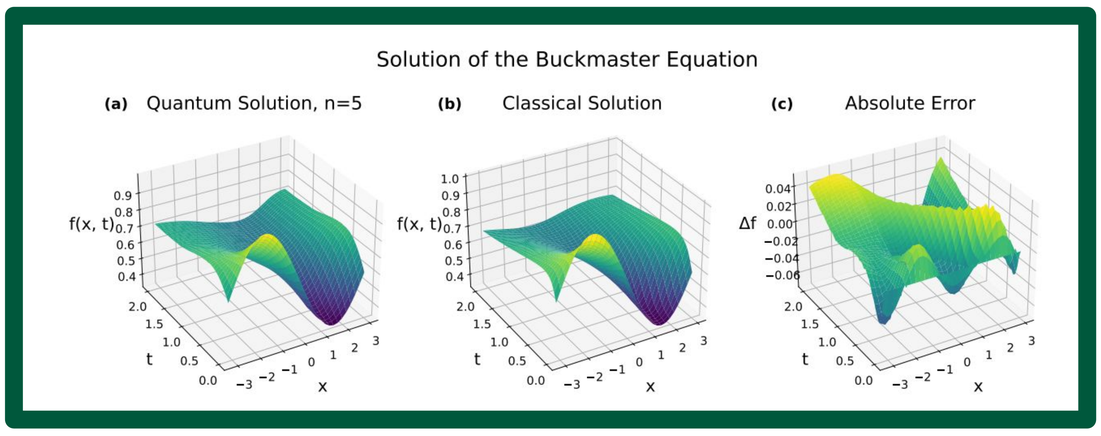
Using Quantum Computing to Solve Diffusion Partial Differential Equations

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MOTIVATION

- Primary application in microplastic diffusion and polymer advantage analysis
- Creating a tool to more efficiently solve partial differential equations (PDE's) that govern microplastic diffusion
- We use Quantum computing to take advantage of the lower computational cost that Quantum Compu tain equations

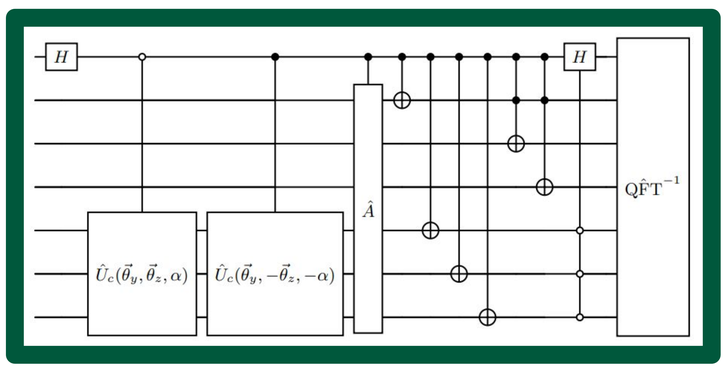
$$\frac{\partial f}{\partial t} = \frac{\partial^2 f^4}{\partial x^2} + \alpha \frac{\partial f^3}{\partial x}$$

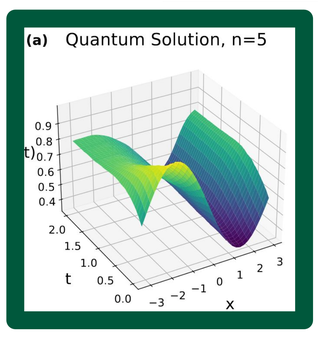


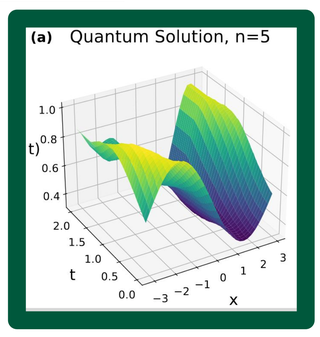
RESULTS

- Simple and known functions were modeled first .
- Reconstructing previous PDE solutions using known Quantum circuits then applying these circuits to increasingly different equations
- We used the Real-Valued ZGR-QFT Ansatz Circuit to solve different functions

$$\frac{\partial f}{\partial t} = \beta \frac{\partial^2 f^4}{\partial x^2} + \alpha \frac{\partial f^3}{\partial x}$$

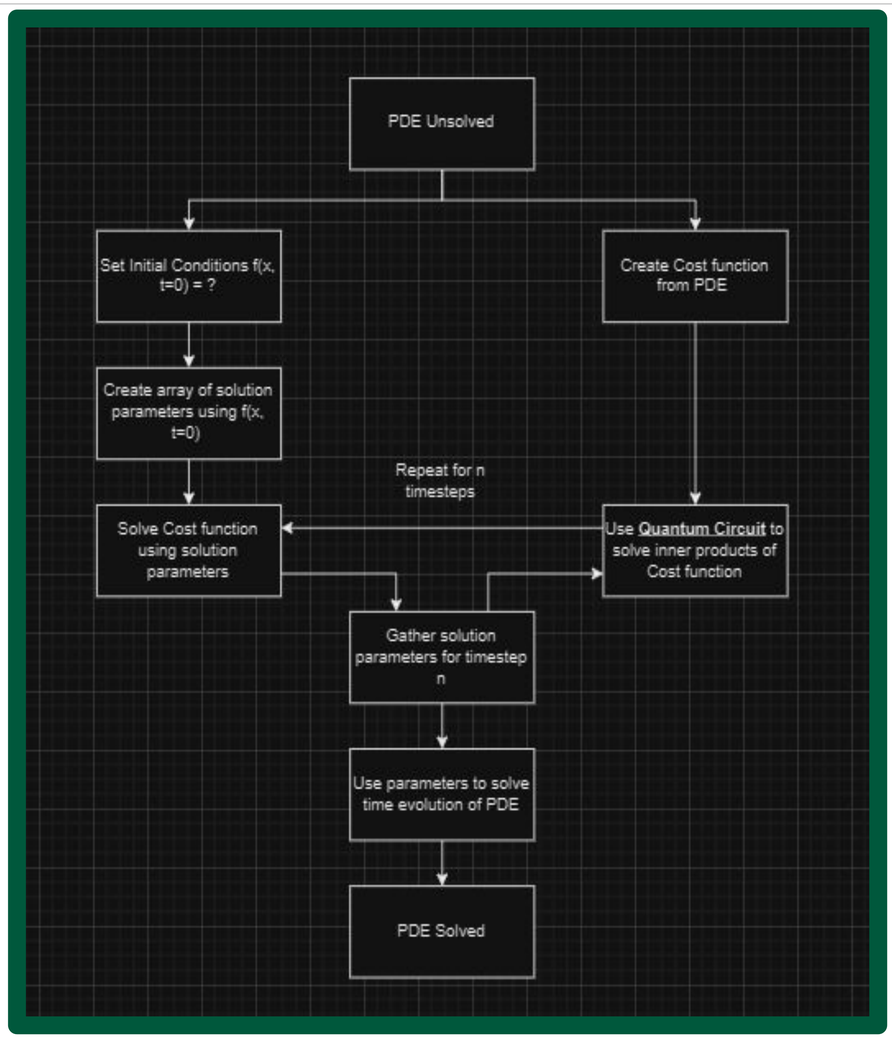






- We added a constant term to the second derivative term called beta which was varied to see the effect of beta on the original equation for beta less than 1
- Applying a beta to the original equation resulted in the graphs on the right using the circuit on the left for increasing beta
- All the graphs were obtained using a budget of 800 with initial and boundary conditions being periodic with 5 qubits and 250 timesteps

WORKFLOW



CONCLUSION

- Using Quantum Computing to solve PDEs provides many logistical advantages computationally such as time to compute and versatility of circuits for solving equations/ simplicity
- Quantum computing is highly specified in its applications possible advantages of solving equations using Quantum computing vary depending on the type of equation you are studying.
- In our case, using a Euler discretization scheme, for a (d+1) dimension PDE, the possible quantum advantage is of the same dimension as M initial data for time T and precision ε with initial data M the comparison of Quantum vs Classical is shown below where r_1 is equal to -4d and r_2 is equal to -9 - 4d.

$$\text{PDE: } \mathcal{O}\left(\frac{\mathcal{C}}{\mathcal{Q}}\right) = \tilde{O}\left(\frac{M}{T^2} d^{r_1} \left(\frac{1}{\epsilon}\right)^{r_2}\right)$$

FUTURE RESEARCH

- Further solving of different equation types can be used to see the flexibility of the program
- Increasing the versatility of this open source program and implement features for plugin equations and solutions
- Further development of the program to work on actual Quantum computers versus simulators on supercomputers
- Further decreasing error in quantum simulations and optimizers used

REFERENCES

- Sarma, Abhijat, Thomas W. Watts, Mudassir Moosa, Yilian Liu, and Peter L. McMahon. "Quantum variational solving of nonlinear and multi-dimensional partial differential equations." *arXiv preprint arXiv:2311.01531* (2023).
- Jin, Shi, and Nana Liu. "Quantum algorithms for computing observables of nonlinear partial differential equations." *arXiv preprint arXiv:2202.07834* (2022).

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